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eAppendix 1: Search Strategies

OID MEDLINE

- 1 Fluid respons*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 2 Fluid Therapy/
- 3 fluid administrat*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 4 fluid resuscit*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 5 cardiac output.mp. or Cardiac Output/
- 6 stroke volume.mp. or Stroke Volume/
- 7 pulse pressure.mp. or Blood Pressure/
- 8 haemodynamic*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 9 hemodynamic*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 10 Hemodynamics/
- 11 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10
- 12 dynamic measur*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 14 passive leg rais*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 14 fluid challeng*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 15 fluid bolus challeng*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 16 Trendelenburg maneuver*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]
- 17 Intrathoracic pressure*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary

concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]

18 12 or 13 or 14 or 15 or 16 or 17

19 septic shock.mp. or Shock, Septic/

20 critical*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]

21 critically ill.mp. or Critical Illness/

22 Sepsis/ or sepsis.mp.

23 Critical Care Nursing/ or Critical Care/ or critical.mp.

24 infect*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]

25 infect*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]

26 infection.mp. or Infections/

27 intensive care.mp.

28 hypotens*.mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms, population supplementary concept word, anatomy supplementary concept word]

29 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28

30 11 and 18 and 29

31 limit 30 to (clinical trial, all or clinical trial, phase i or clinical trial, phase ii or clinical trial, phase iii or clinical trial, phase iv or clinical trial or controlled clinical trial or randomized controlled trial)

32 limit 30 to (adaptive clinical trial or clinical study or clinical trial, all or clinical trial, phase i or clinical trial, phase ii or clinical trial, phase iii or clinical trial, phase iv or clinical trial or comparative study or controlled clinical trial or randomized controlled trial)

EMBASE

1 (fluid respons* or fluid administrat* or fluid resuscit* or cardiac output or stroke volume or pulse pressure or haemodynamic* or hemodynamic*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword heading word, floating subheading word, candidate term word]

2 (dynamic measur* or passive leg rais* or fluid challeng* or fluid bolus challeng* or Trendelenburg maneuver* or Intrathoracic pressure*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword heading word, floating subheading word, candidate term word]

3 (septic shock or sepsis* or critical* or critically ill or infection* or intensive care or hypotens* or resuscitat*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword heading word, floating subheading word, candidate term word]

4 fluid resuscitation/

5 heart output/

6 heart stroke volume/

7 pulse pressure/

8 hemodynamics/

9 blood pressure/
 10 hemodynamic monitoring/
 11 hemodynamic parameters/
 12 hemodynamic stress/
 13 1 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12
 14 dynamics/
 15 fluid therapy/
 16 2 or 14 or 15
 17 septic shock/
 18 critically ill patient/
 19 intensive care/
 20 hypotension/
 21 3 or 17 or 18 or 19 or 20
 22 13 and 16 and 21
 23 limit 22 to (human and (clinical trial or randomized controlled trial or controlled clinical trial or multicenter study or phase 1 clinical trial or phase 2 clinical trial or phase 3 clinical trial or phase 4 clinical trial))

CINAHL

S25	S7 AND S22 AND S23 AND S24
S24	S17 OR S18 OR S19 OR S20 OR S21
S23	S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16
S22	S1 OR S2 OR S3 OR S4 OR S5 OR S6
S21	(MH "intervention trials")
S20	(MH "clinical trials")
S19	(MH "clinical trial")
S18	(MH "randomized controlled trials")
S17	Randomiz* OR Randomis* OR Randomised controlled trial* OR Trial* OR Randomized controlled trial* OR Clinical trial*
S16	(MH "intensive care units")
S15	(MH "infection")
S14	(MH "critically ill patients")
S13	(MH "critically ill patient")
S12	(MH "critical illness")
S11	(MH "critical care")
S10	(MH "sepsis")
S9	(MH "Shock, Septic")

S8 septic shock OR sepsis* OR critical* OR critically ill OR
infect* OR Intensive care OR Hypotens*

S7 "dynamic measur*" OR "passive leg rais*" OR "fluid
challeng*" OR "fluid bolus challeng*" OR "Trendelenburg
maneuver*" OR "Intrathoracic pressure"

S6 (MH "hemodynamics")

S5 (MH "stroke volume")

S4 (MH "cardiac output")

S3 (MH "Resuscitation")

S2 (MH "Fluid Therapy") OR (MH "Fluid Resuscitation") OR
(MH "Fluid Intake")

S1 Fluid respons* OR fluid administrat* OR fluid resuscit*
OR cardiac output OR stroke volume OR pulse pressure
OR haemodynamic* OR hemodynamic*

Cochrane Central Register of Controlled Trials and Cochrane Database of Systematic Review

fluid NEXT respons* OR fluid NEXT administrat* OR fluid NEXT esuscit* OR cardiac NEXT output
OR stroke NEXT volume OR pulse NEXT pressure OR haemodynamic* OR hemodynamic*

AND

dynamic NEXT measur* OR passive NEXT leg NEXT rais* OR fluid NEXT challeng* OR fluid NEXT
bolus NEXT challeng* OR Trendelenburg NEXT maneuver* OR Intrathoracic NEXT pressure*

AND

septic NEXT shock or sepsis* OR critical* OR critically NEXT ill OR infect* OR intensive NEXT care
OR hypotens*

SCOPUS

“fluid respons*” OR “fluid administrat*” OR “fluid resuscit*” OR “cardiac output” OR “stroke volume”
OR “pulse pressure” OR haemodynamic* OR hemodynamic*

AND

“dynamic measur*” OR “passive leg rais*” OR “fluid challeng*” OR “fluid bolus challeng*” OR
“Trendelenburg maneuver*” OR “Intrathoracic pressure*”

AND

“septic shock” or sepsis* OR critical* OR “critically ill” OR infection* OR “intensive care” OR
hypotens*

AND

Randomiz* OR Randomis* OR “Randomised controlled trial*” OR “Trial*” OR “Randomized controlled
trial*” OR “Clinical trial*”

Clinicaltrials.gov

“fluid response” OR “fluid responsiveness” OR “fluid administration” OR “fluid resuscitation” OR “cardiac
output” OR “stroke volume” OR “pulse pressure” OR haemodynamic OR hemodynamic

AND

“dynamic measure” OR “passive leg raise” OR “fluid challenge” OR “fluid bolus challenge” OR
“Trendelenburg maneuver*” OR “Intrathoracic pressure”

AND

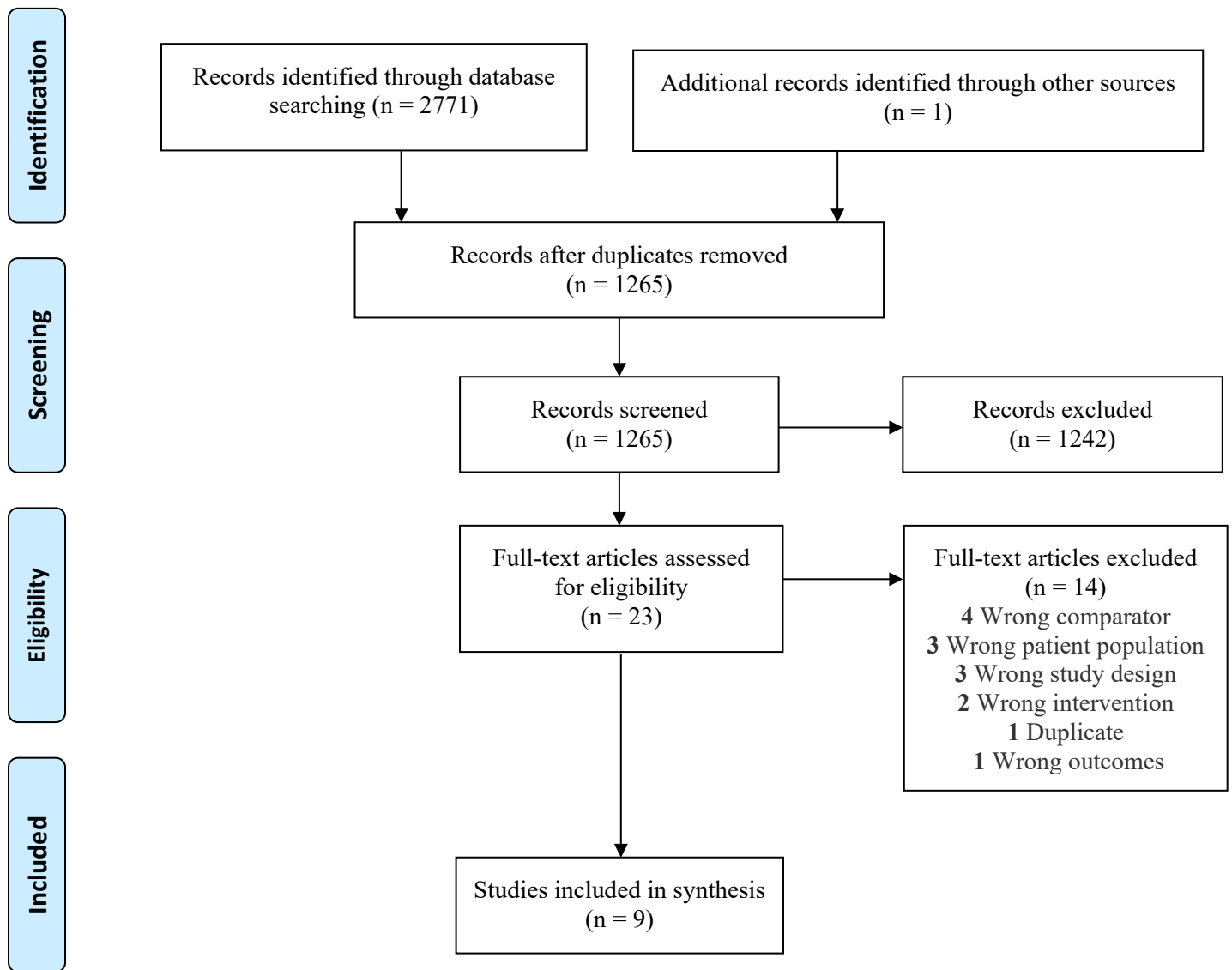
“septic shock” or sepsis* OR critical OR “critically ill” OR infecte OR infection OR “intensive care” OR hypotension OR hypotensive

medRxiv

fluid respons* AND septic shock AND dynamic measure* AND trial*

eTable 1: Certainty Targets for Outcomes

Outcome Hierarchy	Outcomes	Certainty Target
Clinical Outcomes	28-Day Mortality	Difference of 10 per 1000
	ICU-Mortality	Difference of 10 per 1000
	Intensive Care Unit Length of Stay	Difference of 0.5 days
	Hospital Length of Stay	Difference of 1 day
	Requirement for Mechanical Ventilation	Difference of 50 per 1000
	Duration of Mechanical Ventilation	Difference of 0.5 days
	Requirement for RRT	Difference for 25 per 1000
	Acute Kidney Injury	Difference of 50 per 1000
Provision of Resuscitative Therapies	Vasopressor Administration	Difference of 25 per 1000
	Inotrope Administration	Difference of 25 per 1000
	Duration of Vasoactive medications	Difference of 0.5 days
	Total Intravenous Fluid Administration at 24 hours	Difference of 0.5 litres
	Cumulative Fluid Balance at 72 hours	Difference of 1 litre

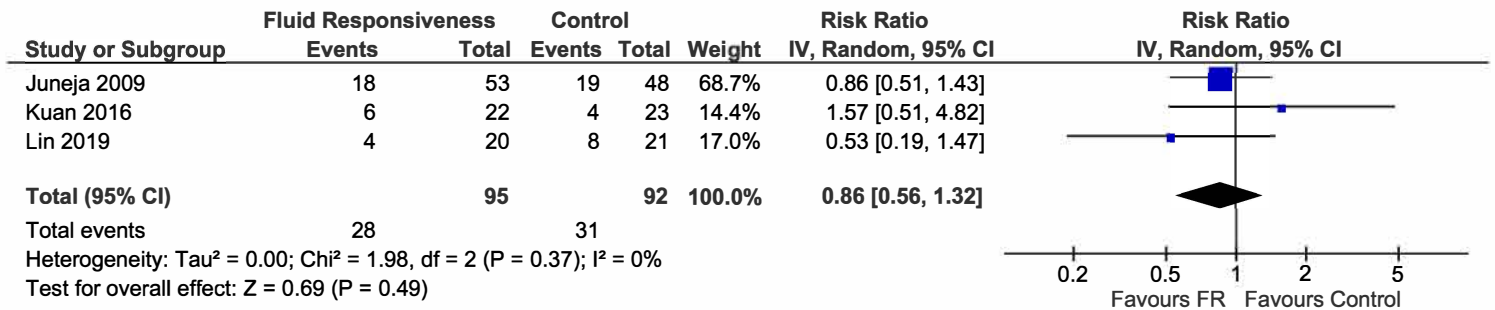


eFigure 1: PRISMA flow diagram. PRISMA indicates preferred reporting items for systematic reviews and meta-analyses.

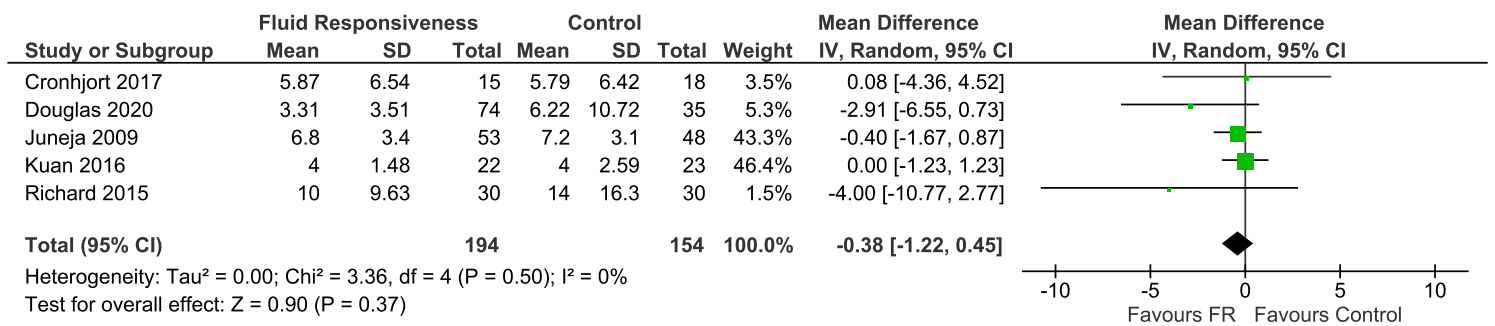
eTable 2: Risk of bias assessment of eligible studies

Study ID	Randomization Process	Deviations from Intended Interventions	Missing Outcome Data	Measurement of Outcome	Selection of Reported Results	Other: Competing Risks Due to Early Termination	Overall
Chen <i>et al</i>, 2015	Probably Low	High	Low	Low	Low	Low	High
Cronhjort <i>et al</i>, 2017	Probably Low	Probably High	Low	Low	Probably High	Low	High
Douglas <i>et al</i>, 2020	Low	High	Low	Low	Low	Low	High
Gruartmoner <i>et al</i>, 2019	Low	Probably High	Low	Low	Low	Low	High
Juneja <i>et al</i>, 2009	Probably Low	Probably High	Probably Low	Low	Low	Low	High
Kuan <i>et al</i>, 2016	Low	Probably Low	Low	Low	Low	Low	Low
Li <i>et al</i>, 2019	Probably low	Probably High	Low	Probably low	Probably Low	Low	High
Lin <i>et al</i>, 2019	Probably low	Probably High	Low	Low	Probably Low	Low	High
Richard <i>et al</i>, 2015	Low	Probably Low	Low	Low	Low	Low	Low

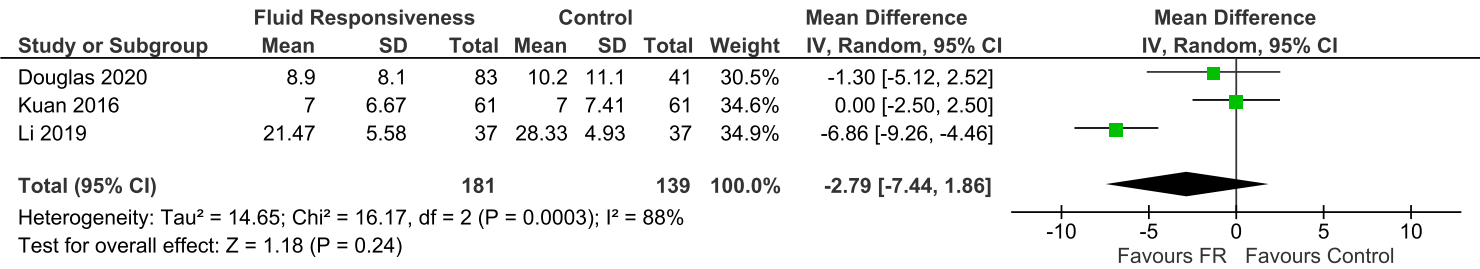
eFigure 2: Forest plot of ICU-Mortality



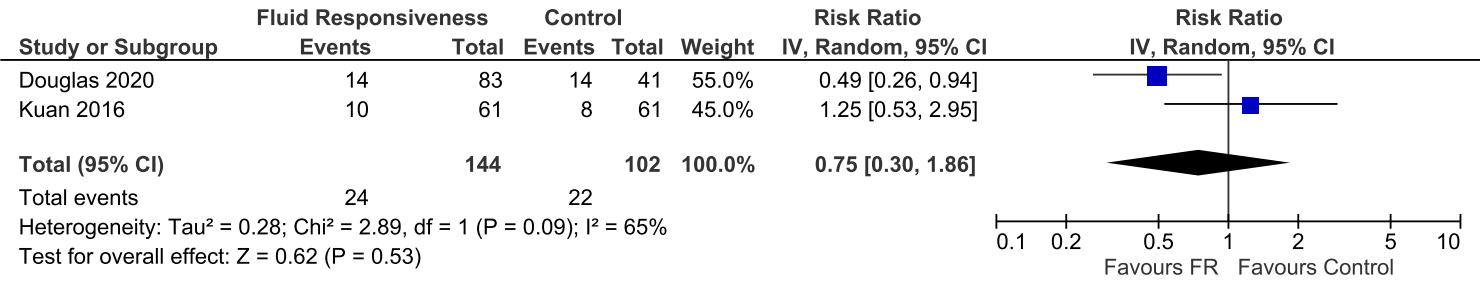
eFigure 3: Forest plot of ICU Length of Stay



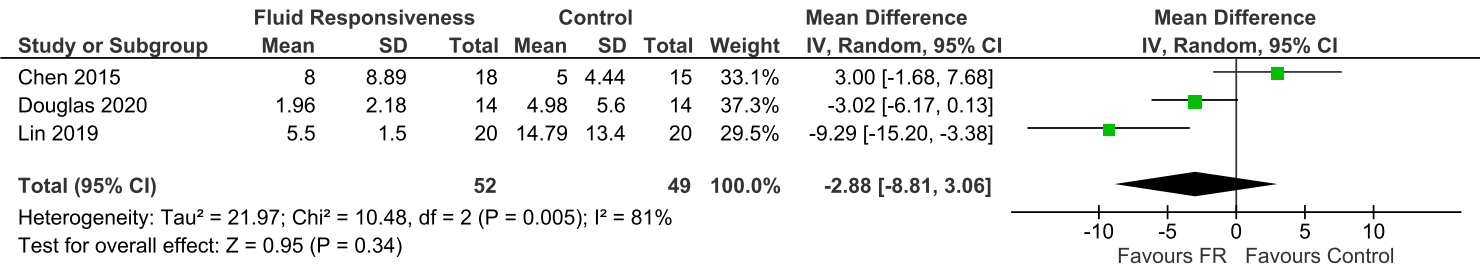
eFigure 4: Forest plot of hospital length of stay



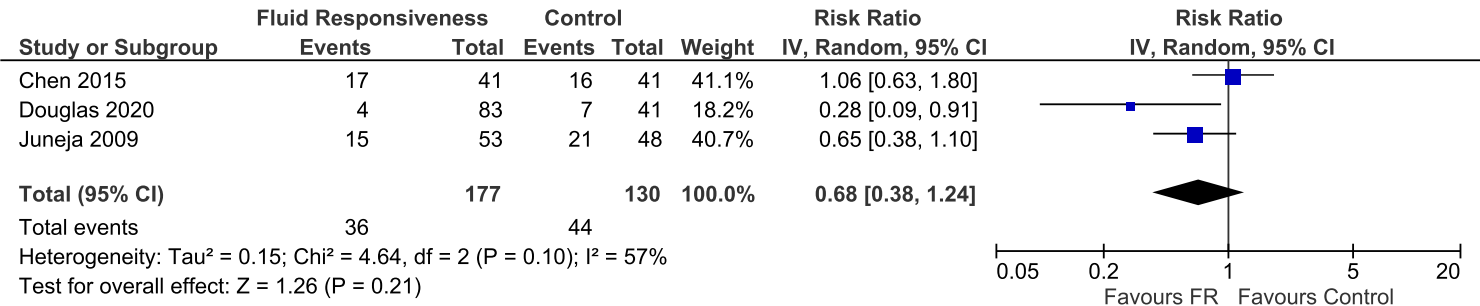
eFigure 5: Forest plot of the provision of mechanical ventilation



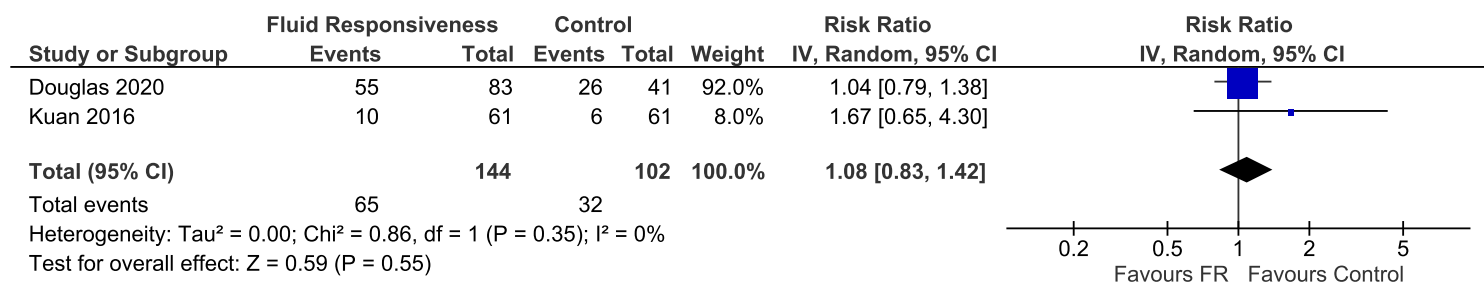
eFigure 6: Forest plot of the duration of mechanical ventilation



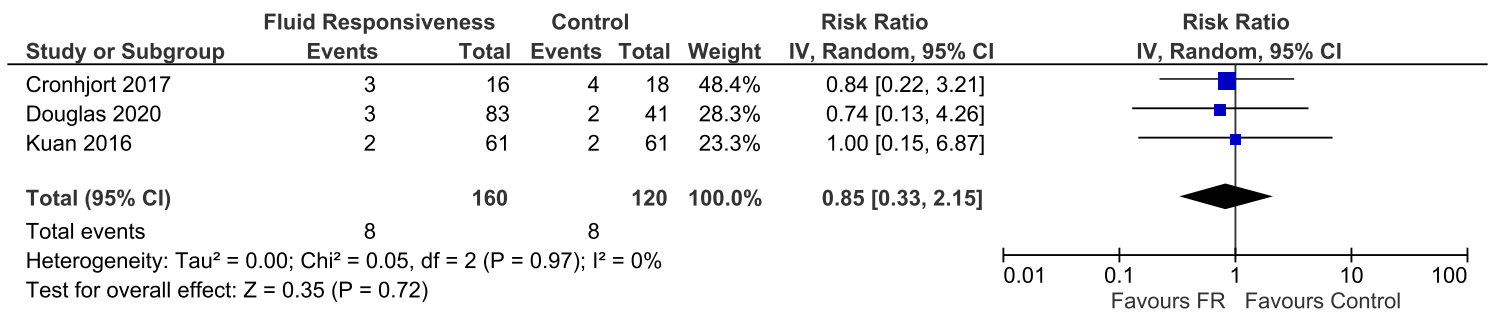
eFigure 7: Forest plot of the provision of renal replacement therapy



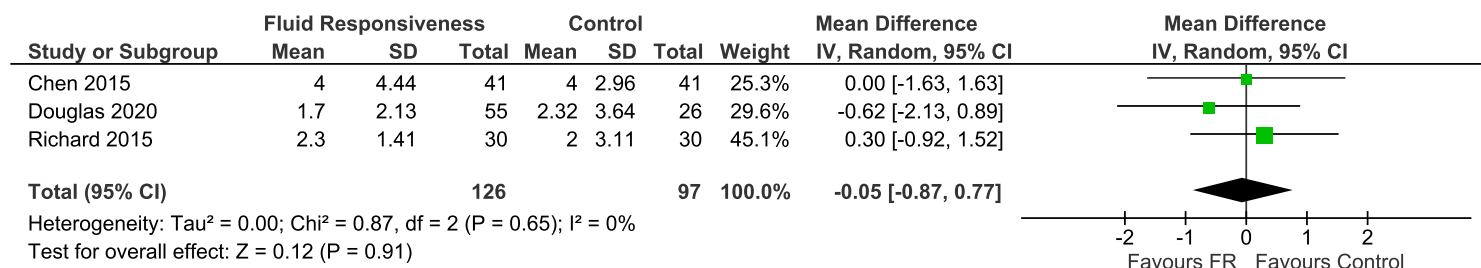
eFigure 8: Forest plot of the administration of vasoactive medications



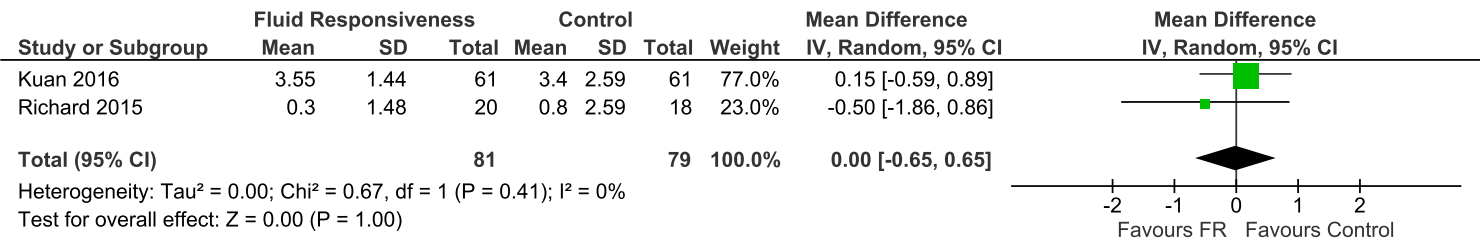
eFigure 9: Forest plot of inotrope administration



eFigure 10: Forest plot of the duration of vasoactive medications



eFigure 11: Forest plot of intravenous fluid administration on day 1





PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	pages 2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	pages 5, 6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	pages 5,6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	page 7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	page 8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	page 8, eAppendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	page 8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	page 9
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	page 8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	page 9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	page 10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	page 9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	pages 9,10
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	page 9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	page 9
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	page 9
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	page 9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	page 10
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	page 10



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	page 11
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	eFigure 1
Study characteristics	17	Cite each included study and present its characteristics.	page 11, Tables 1-2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	eTable 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	pages 11, 12
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	page 11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	page 11-12, Figure 1-2 and eFigures in Appendix
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	page 11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Tables 3-4
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	pages 11, 12
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	page 13
	23b	Discuss any limitations of the evidence included in the review.	page 13, 14
	23c	Discuss any limitations of the review processes used.	page 13, 14
	23d	Discuss implications of the results for practice, policy, and future research.	page 14-16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	page 7
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	page 7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	page 1
Competing interests	26	Declare any competing interests of review authors.	page 1
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA